

Beyond the Blackout: Qualitative Insights and Sentiment Analysis on the Impact, Adaptation, and Design Solutions During the Internet Shutdown in Bangladesh

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Abstract

The recent period of Internet restriction in Bangladesh, which occurred from July 14 to August 5 during the anti-discrimination student movement, had a significant impact on individuals' lives. This study aims to know how their communication, information access, and daily activities were affected and how they used to cope with that, again what were their expectations from social media and communication tools. Through interviews with 15 individuals from diverse demographics, supported by qualitative analysis and sentiment analysis, the findings provide valuable insights into the challenges faced, experiences shared, and recommendations offered by participants. Based on these findings, we propose design implications to develop a robust digital infrastructure capable of addressing similar challenges in future scenarios. Our work contributes to CHI by serving as a foundation for policymakers and designers to create effective solutions that mitigate the adverse effects of Internet shutdowns.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Sentiment Analysis**;

Keywords

Qualitative Studies, Human-Computer Interaction

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1 Introduction

The recent Internet restriction and shutdown in Bangladesh, from July 14 to August 5, left the country disconnected from global connectivity for nearly five days during the anti-discrimination student movement. These restrictions severely affected communication, information access, and daily activities, highlighting the critical role of Internet in modern life. The shutdown created unique challenges for individuals, forcing them to develop coping mechanisms and adapt to limited connectivity while also exposing design shortcomings in existing digital platforms. To investigate these impacts, this study focuses on addressing the following research questions:

- **RQ1:** What are the impacts of the Internet shutdown on individuals' communication, information processing, decision-making, and daily lives?
- **RQ2:** How do sentiments vary during the Internet shutdown based on individuals' experiences?
- **RQ3:** How do individuals adapt to the Internet shutdown, and what roles do alternative communication, information sources, and community networks play in decision-making and socio-political awareness?
- **RQ4:** What design features and improvements in social media apps and devices could have enhanced user experiences, safety, and adaptability during the Internet shutdown and movement period?

After interviewing 15 individuals from different demographics, we conducted a qualitative study and analyzed their sentiments using various models. Findings reveal the impacts of the shutdown on

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individuals' lives, their emotions regarding the crisis, their adaptive strategies, and their expectations from social media and communication tools. Based on these findings, we propose design implications to enhance digital systems for better resilience during crises. Our work contributes to CHI by providing insights, which can help policymakers and designers create solutions that better address the challenges faced during future Internet shutdowns.

2 Background

The July 2024 Internet shutdown in Bangladesh was precipitated by widespread protests following Sheikh Hasina's controversial fourth-term election victory, extending her 15-year rule amid allegations of human rights violations. On July 1, the Anti-Discrimination Student Movement launched protests demanding reforms to civil service reservation quotas, marking the beginning of a significant social outburst [11]. Tensions escalated on July 14 when Hasina made inflammatory remarks about the protesters at a press conference, further fueling nationwide agitation [33]. By July 15, the BTRC ordered 3G and 4G Internet shutdowns in Shahbag and key protest sites, later expanding to universities. Violence erupted when Chhatra League activists attacked quota protesters at Dhaka University, and images of injured female students circulated widely on social media, sparking nationwide condemnation. The killing of Abu Sayed, a student at Begum Rokeya University, by police on July 16 further intensified protests, prompting the BTRC to extend the shutdown to 59 universities and 205 sites nationwide [34].

On July 18, the NTMC banned Facebook and YouTube, restricted 4G to 2G, and the BTRC falsely blamed Internet disruptions on Mohakhali data center fires, a narrative debunked by journalists from The Daily Star, who found no evidence of fire or damage at the sites [32]. Broadband services were partially restored on July 23 and fully reinstated by July 24, but mobile Internet remained restricted until July 28, with social media and communication apps blocked. Protests continued into August, leading the NTMC to block Telegram, Messenger, and Facebook on August 2. On August 4, 4G services were suspended, and broadband was shut down on August 5. However, all Internet services were restored on August 5 when Sheikh Hasina resigned and left the country [34].

3 Related Work

Numerous studies have explored the impacts of Internet shutdowns [10, 26, 31, 37], individuals' coping strategies [8, 23, 35], and design recommendations [2, 6]. Grandhi et al. examined perceptions of long-term shutdowns, highlighting personal and societal impacts with implications for preparedness and design [14]. Lupien et al. investigated Internet dependency and responses to shutdowns, revealing mixed effects on work, knowledge, and social ties, alongside offline coping strategies, with implications for education and emergency management [22].

Additionally, studies focused on shutdowns during social movements [15, 36]. Guntrum et al. analyzed ICT's role in Myanmar's anti-military protests, examining its effects during shutdowns and proposing solutions like offline applications and secure VPNs [16]. Similarly, Awwad et al. explored how Palestinian activists adapted to shutdowns amid censorship and intimidation, emphasizing low-tech, labor-intensive methods used by Hamas and the Palestinian

Authority [5].

While extensive research has been conducted on Internet shutdowns, their impacts, and potential design solutions during both routine periods and social movements, this study is the first to explore these dynamics in the context of the recent shutdown in Bangladesh.

4 Methodology

The methodology includes questionnaire preparation, interview conduction, transcription, organization, and analysis. A semi-structured questionnaire was developed using existing literature, refined through three pilot interviews, which were conducted online, and finalized with open-ended and Likert-scale questions. Fifteen interviews were conducted in total, including the pilot interviews (12 online via Google Meet, and 3 in-person), with recordings processed using OBS Studio¹ and phone recorders. Transcriptions were refined using TurboScribe² and Whisper-large-V3³, then categorized in a Google spreadsheet for analysis.

To capture the diverse experience of participants, we employed purposive sampling to recruit 15 participants in multiple demographic dimensions: gender, age, place of residence, and level of activity. Of the 15 interviewees, 11 identified as male and 4 as female, 10 lived in Dhaka and 5 lived outside, and 7 were active protesters and 8 were passive. The age brackets were: 18–24 (6), 25–30 (4), 31–40 (3), 41–50 (1), 50+ (1). As the Internet shutdown and subsequent government crackdown were in response to a student led movement, we prioritized the broader student community and recent graduates in order to capture the distinct experiences and perspectives of those most directly affected by these measures.

5 Data Analysis

We employed two different methods for data analysis: thematic analysis and sentiment analysis. Thematic analysis provided insights into the overall impact, adaptation strategies, and design implications, addressing RQ1, RQ3, and RQ4. Meanwhile, sentiment analysis revealed how sentiments have fluctuated during the Internet shutdown, addressing RQ2.

We conducted thematic analysis using Braun and Clarke's six-phase framework [7], streamlining it into four key phases. The interviews were transcribed and systematically reviewed, with key excerpts highlighted and categorized into a structured dataset. Using an inductive coding approach [9], we generated 231 distinct themes, appearing 901 times across sections. These themes were refined iteratively, with senior authors reviewing, merging, and reorganizing them to ensure clarity and coherence. This rigorous process allowed a structured interpretation of the participant experiences, which formed the foundation for our qualitative insights.

Prior works on Sentiment Analysis used various machine learning techniques [3, 18, 20, 24, 27, 38, 40], as well as various transformer and LLM approach [17, 19, 21, 25, 28, 30, 40] and tools [12]. Sentiment analysis involved selecting suitable emotion classification models as indicated in Table 1, integrating them into an analysis pipeline, and systematically applying them to each dataset entry.

¹<https://obsproject.com/>

²<https://turboscribe.ai/>

³<https://huggingface.co/openai/whisper-large-v3>

The transformer library pipeline facilitated model execution, identifying the top three emotions per entry. Results were visualized through graphs, aggregating the top emotions for each participant and question into a total score, offering a clear overview of emotional patterns throughout the interview.

6 Findings

This section highlights the participants' experiences, the impacts of the shutdown on their day-to-day decision making and information processing, and how they adapted with those. Moreover, this section also includes the recommendations and expectations of social media apps and devices from them.

6.1 Impacts of the Internet Shutdown on Information Processing, Decision Making and Daily Life

Our findings revealed important aspects of the impact of the Internet shutdown on individual's life, particularly focusing on communication, information-processing and daily life.

6.1.1 Impact of the Internet Shutdown on Communication and Information Processing. The Internet shutdown had profound consequences on communication, disrupting critical social and economic interactions. Most of the participants reported their struggle to communicate with near and dear ones to get their important information. P13 conveyed a sense of being in the dark due to a lack of communication:

"90% of the time, we were in the dark. I struggled to get reliable updates about my cousin, who was actively involved in the movement. Every 2-3 days, I somehow managed to call him to check on his well-being and find out what was happening, but those calls were neither smooth nor dependable." (P13)

Many participants reported significant challenges in acquiring and verifying information during the curfews and Internet shutdowns. Limited access to trusted sources and delayed information severely impacted their ability to stay informed. Although a few highlighted the reliability of basic news from selective channels, notably Jamuna TV, significant distrust arose due to perceived government influence and propaganda, leading to widespread skepticism and rejection. For instance, P8 said:

"I've always been skeptical about TV news. I feel they're always biased towards the ruling party... During the movement, my skepticism increased significantly, and I stopped believing them at all. For instance, Somoy TV often published utterly fabricated news in favor of the ruling party, Awami League. Other channels avoided covering important news that reflected poorly on the ruling party, like updates on the casualties—how many children and young adults were killed by the police under government orders." (P8)

6.1.2 Impact of the Internet Shutdown on Decision-Making. The absence of reliable information during the shutdown profoundly disrupted decision-making of the individuals. A portion of the participants expressed difficulty trusting available information sources

and felt heavily dependent on a limited number of unreliable options, further complicating their decision-making. Again, some participants reported that their decision-making was marked by rushed and tough choices under restricted circumstances, often driven by incomplete or unreliable information. P5 recounted a narrative from his life to illustrate the severity:

"At a tea shop near our house, people were discussing how we wouldn't be able to buy essentials starting the next day. Panic started to spread fast, and everyone rushed out to stock up on whatever they could—potatoes, eggs, rice. The atmosphere was tense, and the scramble for supplies felt surreal." (P5)

6.1.3 Impacts of the Internet Shutdown on Daily Life. All participants reported significant disruptions to their daily lives and social interactions due to the inability to access timely and reliable information, which affected their ability to communicate, access resources, and maintain normal routines. A few respondents experienced tangible challenges, such as difficulties in accessing essential resources like food, medicine, or financial services, due to the lack of updates about the safety around their surroundings. Access to healthcare was disrupted for some, with participants citing difficulties in acquiring medical assistance and information during the shutdown.

6.2 Sentiment Variations During the Shutdown

Employing the four different models as stated above, varying sentiments were detected from distinct people. As shown in Figure 1, model 1 detected "sadness" as the dominant emotion in 8 individuals, "joy" in 5 individuals, and "fear" in 2 others. This clearly shows how the shutdown resulted in significant frustration among individuals, and when they were talking about their experiences, their sorrow was expressed. Again, some individuals had joy in their expression because the protest was successful, and they did not lose any of their loved ones. Moreover, some expressed fear regarding discussions of the traumatic experience throughout the shutdown in their lives.

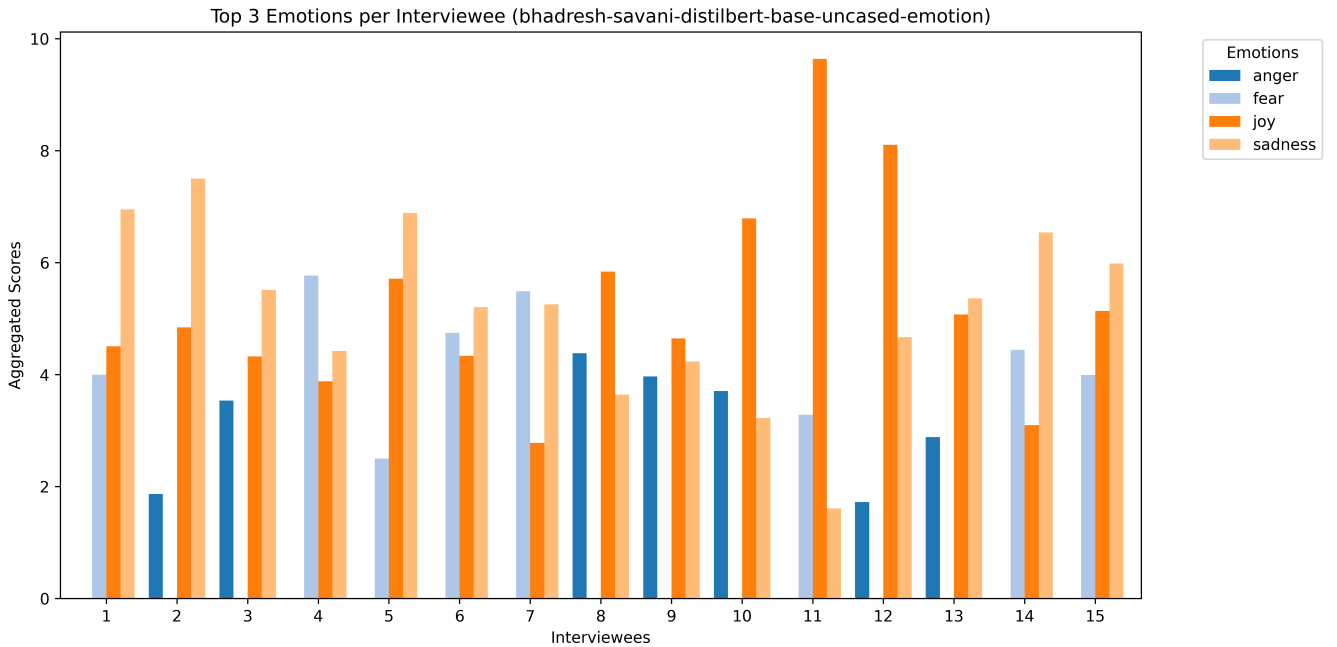
Similarly, as shown in Figure 2, model 2 identified "neutral" as the dominant emotion in 10 individuals and "fear" as the top emotion in 5 individuals. This could be attributed to the predominantly informational nature of the questions, which prompted responses with a neutral tone. Consequently, as shown in Figure 3 and Figure 4, models 3 and 4 performed poorly, as they classified all or most responses as "neutral," failing to capture the underlying emotional context.

6.3 Adaptation Strategies and the Role of Networks and Information Sources

Our findings revealed various coping mechanisms employed by individuals to address communication and access challenges, multimedia sharing and bandwidth limitations, as well as the role of networks in community support and risk management

6.3.1 Adapting Communication and Access Strategies During Restrictions. The responses highlighted diverse strategies for adapting to communication limitations, with phone calls, SMS, and VPNs being the most utilized. During the complete Internet shutdown,

No	Model Name	Description
1	bhadresh-savani/distilbert-base-uncased-emotion ⁴	This model is a fine-tuned version of DistilBERT-Base-Uncased ⁵ model, trained using the dataset from the Twitter-Sentiment-Analysis ⁶ dataset.
2	j-hartmann/emotion-english-distilroberta-base ⁷	This model is fine-tuned from DistilRoBERTa-Base ⁸ model using six datasets for emotion detection. It can identify seven types of emotions: anger, disgust, fear, joy, sadness, surprise, and neutral.
3	michellejieli/emotion_text_classifier ⁹	This model is fine-tuned from the model-2 to classify seven emotion categories using a dataset derived from the transcripts of the television show F.R.I.E.N.D.S..
4	SamLowe/roberta-base-go_emotions ¹⁰	This model is fine-tuned from RoBERTa-Base ¹¹ using the GoEmotions ¹² dataset. It supports multi-label classification for emotion detection, identifying a total of 28 emotions.

Table 1: Descriptions of Used Models for Sentiment Analysis**Figure 1: Sentiment Analysis of the Experiences of the Interviewees Using Model 1**

reliance on phone calls and SMS emerged as the primary method of communication. Despite phone calls being perceived as largely ineffective due to high call rates, poor network quality, and the inability to share multimedia, and SMS being limited by high costs, character restrictions, and outdated functionality, phone communication remained an essential means of maintaining connectivity. Interviewee P6 expressed this dual nature of phone communication:

"Using SMS was pretty annoying, but it allowed us to reliably convey information. Given the blackout situation, even this basic utility felt like a blessing. And while there was concrete evidence that they restricted the mobile network too, they didn't shut it off completely. So, we relied on phone calls to stay connected." (P6)

After phone communication, VPNs were mostly used for Internet access during bandwidth restrictions. While many interviewees highlighted challenges like unreliable connections, slow speeds, and limited functionality, which made VPNs ineffective for communication, some interviewees expressed satisfaction with their performance under a mass restriction of bandwidth. Especially the participants who used premium versions of some VPNs, expressed higher level of satisfaction. For instance, P4 said:

"VPN was the best workaround. I mean, I was able to access the Internet at least, be it at a slow speed. I was able to see the people I love, I could text them, I was able to read the news. The use experience was not great, but it was the best we could have had back then." (P4)

Furthermore, some technically skilled individuals set up local servers using alternative networks, and some utilized the BDIX[1] server, a data exchange system among Internet service providers in Bangladesh that operates independently of international submarine cables, to create locally hosted servers for sharing media and communicate. These servers played a vital role in facilitating connections among people.

6.3.2 Coping with Multimedia Limitations and Bandwidth Challenges. Although phone-based communication allowed minimal functionality, participants were frustrated by bandwidth limitations that made sharing multimedia nearly impossible. Some individuals could share multimedia using VPNs, but this method didn't always work well, and was not universally reliable. Again, some participants adapted by using locally hosted or BDIX servers, but their utility was limited by reliance on the server host being online and running, as well as low speeds in certain cases. This led many to completely cease multimedia sharing activities. The psychological impact of restrictions also played a role in hindering adaptation. Some participants reported that they were unable to adapt and investigate alternative solutions due to psychological factors, such as feelings of vulnerability and fear of repercussions. P4 expressed his fear in multimedia sharing:

"Additionally, there was a constant fear of sharing multimedia because of the risk posed by corrupt law enforcement and government-backed intelligence or monitoring groups. This fear made me hesitant to even try, further hampering my ability to connect with others or stay informed about the movement." (P4)

6.3.3 Role of Networks in Community Support and Risk Management. Although a notable portion of participants reported no involvement in or awareness of any ad-hoc communication network, many actively participated in or benefited from such community networks. Some communities lacked pre-existing networks, but they spontaneously formed during the movement period in response to the necessity for communication and coordination, thereby illustrating the adaptive nature of social networks in action. The community network was a source of assistance for certain participants. For example, P15 expressed her appreciation:

"The online networks and community groups were my lifeline. They kept me updated about everything—the truth about what was happening. Without them, I would have been completely lost." (P15)

Community networks played a significant role in spreading news in real-time, highlighting their importance in ensuring timely updates, maintaining situational awareness, mitigating risk, and saving lives. P9 expressed how community networks saved her life:

"I decided to join the physical protest one day. But my friends warned me that three young people had been killed an hour earlier at the spot I was heading to. I waited for an hour to avoid the life risk and joined later when the protest regained momentum and got stronger. So you can see that if it weren't for this network or group, I may as well could have been one of those three unfortunate protesters." (P9)

However, some respondents highlighted certain issues of community networks, such as the lack of authenticity in shared information and the limited effectiveness of these groups in addressing critical needs.

6.4 Expectations from Social Media Apps and Devices

Given the impacts of the Internet shutdown and the coping mechanisms employed by individuals, participants highlighted several design shortcomings in social media apps and devices, expressing a strong need for targeted improvements. Flexible connectivity solutions, such as apps capable of seamlessly switching between networks (e.g., mobile data, Wi-Fi, or peer-to-peer), were seen as vital for maintaining communication. P11 suggested integrating peer-to-peer functionality into widely used platforms like Messenger to improve accessibility during shutdowns, noting:

"Not many people are familiar with platforms like Bridgify or Briar, so switching to those platforms suddenly might not be feasible, even though they are good solutions. Since Messenger is widely used in the country, an additional background architecture enabling peer-to-peer communication could go a long way in helping people tackle situations like Internet shutdowns." (P11)

Participants also emphasized efficient data management through adjustable multimedia compression settings, lightweight app versions, and low-bandwidth servers optimized for emergencies. To address security concerns, participants proposed secure cloud backups for rapid local deletion and data recovery, second-profile features to display only non-sensitive content during inspections, and hidden input systems to redirect unauthorized access to decoy pages. Biometric-triggered emergency modes were also suggested for instantly concealing sensitive content. Additionally, localized emergency alerts tailored for crises were recommended to ensure timely updates.

Together, these recommendations underscore the need for adaptable and secure app functionalities to enhance resilience during crises.

7 Discussion

This study shows how Internet shutdowns affect communication, decision-making, and daily life, emphasizing the importance of connectivity in modern society.

7.1 Impacts of the Shutdown and the Challenges Faced for it

The internet shutdown exposed severe inefficiencies in social media platforms, particularly Facebook and Messenger, which failed to support even basic text exchanges, leaving users disconnected. VPNs provided limited access, but messages failed to send, and multimedia sharing was impossible, worsening the crisis. The government claimed the blackout resulted from the destruction of data centers and fiber-optic cables [4], but post-movement findings confirmed it was a deliberate act to suppress information [13]. Government surveillance increased fear, with many suspecting their calls and messages were monitored, leading to self-censorship.

and avoidance of discussions on protests and crackdowns. With social media and online news blocked, mainstream state-aligned media dominated, spreading filtered narratives and deepening public distrust. Phone calls and SMS became the only alternatives, but high costs, character limits, and restrictions on foreign calls hindered communication. Similar measures restricting internet access have been observed in Iran, Myanmar, China, and Russia [15][16][39][29].

The lack of fact-checking fueled misinformation, as people relied on personal networks and word-of-mouth, often spreading unverified rumors—a pattern also seen in prior research [15]. Curfews further restricted movement, disrupting healthcare, emergency services, and businesses, particularly affecting freelancers and contractual workers relying on the internet. Tensions within families grew, as younger individuals accessed VPNs for accurate information, while older members relied on biased TV news, a divide also noted in Iran [15].

The shutdown exposed critical weaknesses in Bangladesh’s digital infrastructure, highlighting the urgent need for policy reforms and resilient communication systems, discussed in the following sections.

7.2 Adaptive Strategies

In response to the internet shutdown, people turned to phone calls and SMS, though high costs, character limits, and congestion made them unreliable. VPNs provided limited access, but during complete shutdown, they became completely unusable.

Citizen journalism became a key information source, with updates shared via SMS and word-of-mouth, though concerns about misinformation and manipulation remained. Fearing government surveillance, people adopted self-censorship, used secondary profiles on their smartphones, or shifted to secondary phones or button phones, and some even pretended to support the government to avoid suspicion. This finding is consistent with previous work [5, 15].

Locally, ad-hoc networks and local servers hosted by tech-savvy individuals using BDIX [1] facilitated information exchange, with spontaneous community watch groups emerging to coordinate efforts. While these networks provided crucial updates, their effectiveness was limited by concerns about credibility and reach. These adaptations highlighted both resilience and the urgent need for more robust communication infrastructures.

7.3 Actionable Design Recommendations and Policy Guidelines for Improved Digital Infrastructure

Based on participants’ experiences and identified shortcomings, we propose the following recommendations to enhance digital resilience during connectivity disruptions.

7.3.1 Flexible Connectivity Solutions. To prevent total communication breakdown, we recommend popularizing *peer-to-peer network based communication apps* like Briar and Bridgify to enable offline messaging. Early government support is essential for preparedness. We also recommend developing *hybrid-architecture* based adaptive

versions of Messenger and WhatsApp, that switch between centralized and peer-to-peer networks for uninterrupted communication [16].

7.3.2 Efficient Data and Bandwidth Management. Participants faced severe application failures under restricted bandwidth. We propose *dynamic compression*, i.e., integrating adaptive compression for media to optimize transmission under network constraints. We also propose creating *lightweight, minimal versions* of messaging apps with core features for low-bandwidth use. There should also be dedicated low-bandwidth servers to sustain critical communication during shutdowns.

7.3.3 Privacy and Security Features. Concerns over surveillance and forced phone inspections highlight the need for *secure remote storages*, i.e., encrypted cloud backups for rapid local content deletion while preserving remote access. We also recommend *compartmentalized profiles*, multiple digital identities with adjustable privacy settings. Our recommendations also include *concealed conversations*, which are hidden chat folders with decoy access pages for security under coercion. We also propose *biometric emergency triggers*, i.e., discreet biometric shortcuts to lock or hide sensitive content instantly.

7.3.4 Alternative Communication Channels. To enable messaging without relying on mobile networks, we suggest *proximity-based messaging*, i.e., bluetooth and NFC-based offline peer-to-peer messaging. We also recommend *decentralized risk notifications* like, community-driven security alerts without internet dependency.

7.3.5 Other Policy-Level Recommendations. The government should consider a policy level implementation of *distributed internet architecture*, transitioning from a highly centralized internet model to a controlled distributed network that balances preventing shutdowns while mitigating cyber risks. Policies must also *prevent mobile operators from exploiting crises by drastically increasing call and SMS rates* during times of heightened demand.

By implementing these recommendations, digital infrastructure can become more resilient, secure, and accessible during future disruptions, ensuring continued communication and information flow.

8 Conclusion and Future Work

This study explores the impact, adaptation, and design solutions during the recent Internet shutdown in Bangladesh. By taking interviews from fifteen individuals and doing qualitative study and sentiment analysis from the responses, the findings highlight how the recent Internet shutdown in Bangladesh affected individuals’ emotions, their access to information in their daily lives, their varying emotions regarding the crisis, their coping mechanisms during the disruption, and their experiences with the existing social media platforms and apps. By addressing their expectations, we provide specific design implications for future systems that are anticipated to enhance assistance for persons during Internet shutdowns. Future research will expand the scope by including more participants, exploring additional dimensions of the issue, and further refining the proposed solutions to enhance their effectiveness.

References

- [1] 2025. *Bangladesh Internet Exchange (BDIX)*. <https://bdix.net/> Accessed: 2025-01-23.
- [2] Andrei Akhremenko, Sergey Zhiglov, Alexander Petrov, and Aleksei Turobov. 2022. The Internet Shutdown during the Protest: a Model of Changing the Network Structure with an Adjustable Level of Continuity of Connections. In *2022 15th International Conference Management of large-scale system development (MLSD)*. IEEE, 1–5.
- [3] Asma Ameer, Sana Hamdi, and Sadok Ben Yahia. 2023. Sentiment Analysis for Hotel Reviews: A Systematic Literature Review. *ACM Comput. Surv.* 56, 2, Article 51 (Sept. 2023), 38 pages. doi:10.1145/3605152
- [4] TBS Report 06 August and TBS Report. 2024. Bizarre claims and apologies: The statements of palak during internet blackouts. <https://www.tbsnews.net/features/panorama/bizarre-claims-and-apologies-statements-palak-during-internet-blackouts-910616> Accessed: 2025-03-07.
- [5] Ghadeer Awwad and Kentaro Toyama. 2024. Digital Repression in Palestine. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15.
- [6] Zachary S. Bischof, Kennedy Pitcher, Esteban Carisimo, Amanda Meng, Rafael Bezerra Nunes, Ramakrishna Padmanabhan, Margaret E. Roberts, Alex C. Snoren, and Alberto Dainotti. 2023. Destination Unreachable: Characterizing Internet Outages and Shutdowns. In *Proceedings of the ACM SIGCOMM 2023 Conference* (New York, NY, USA) (ACM SIGCOMM '23). Association for Computing Machinery, New York, NY, USA, 608–621. doi:10.1145/3603269.3604883
- [7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3 (01 2006), 77–101. doi:10.1191/1478088706qp0630a
- [8] Monica A Bustinza and Kaila Witkowski. 2022. Immigrants, deviants, and drug users: A rhetorical analysis of President Trump's fear-driven tweets during the 2019 government shutdown. *Policy & Internet* 14, 4 (2022), 788–806.
- [9] David Byrne. 2022. A worked example of Braun and Clarke's approach to reflexive thematic analysis. *Quality & quantity* 56, 3 (2022), 1391–1412.
- [10] Michael Collyer and Joss Wright. 2021. A Bayesian Analysis of Collective Action and Internet Shutdowns in India. In *Proceedings of the 13th ACM Web Science Conference 2021* (Virtual Event, United Kingdom) (WebSci '21). Association for Computing Machinery, New York, NY, USA, 309–318. doi:10.1145/3447535.3462480
- [11] Wikipedia contributors. 2024. Anti-discrimination Students Movement. https://en.wikipedia.org/wiki/Anti-discrimination_Students_Movement#:~:text=Anti%2Ddiscrimination%20Students%20Movement%20or,student%2Dled%20quota%20reform%20movement Accessed: 2025-01-20.
- [12] Daniel Coutinho, Luisa Cito, Maria Vitória Lima, Beatriz Arantes, Juliana Alves Pereira, Johny Arriel, João Godinho, Vinicius Martins, Paulo Vitor C. F. Libório, Leonardo Leite, Alessandro Garcia, Wesley K. G. Assunção, Igor Steinmacher, Augusto Baffa, and Balduino Fonseca. 2024. "Looks Good To Me :-)": Assessing Sentiment Analysis Tools for Pull Request Discussions. In *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering* (Salerno, Italy) (EASE '24). Association for Computing Machinery, New York, NY, USA, 211–221. doi:10.1145/3661167.3661189
- [13] freedomhouse.org. 2024. <https://freedomhouse.org/country/bangladesh/freedom-net/2024> Accessed: 2025-03-07.
- [14] Sukeshini A Grandhi, Linda Plotnick, and Starr Roxanne Hiltz. 2020. An internet-less world? Expected impacts of a complete internet outage with implications for preparedness and design. *Proceedings of the ACM on human-computer interaction* 4, GROUP (2020), 1–24.
- [15] Margarita Grinko, Sarvin Qalandar, Dave Randall, and Volker Wulf. 2022. Nationalizing the Internet to Break a Protest Movement: Internet Shutdown and Counter-Appropriation in Iran of Late 2019. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 314 (Nov. 2022), 21 pages. doi:10.1145/3555205
- [16] Laura Gianna Guntrum. 2024. Keyboard Fighters: The Use of ICTs by Activists in Times of Military Coup in Myanmar. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19.
- [17] Mickel Hoang, Oskar Alija Bihorac, and Jacobo Rouces. 2019. Aspect-based sentiment analysis using bert. In *Proceedings of the 22nd nordic conference on computational linguistics*. 187–196.
- [18] Praphula Kumar Jain, Vijayalakshmi Saravanan, and Rajendra Pamula. 2021. A Hybrid CNN-LSTM: A Deep Learning Approach for Consumer Sentiment Analysis Using Qualitative User-Generated Contents. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 20, 5, Article 84 (July 2021), 15 pages. doi:10.1145/3457206
- [19] Archa Joshy and Sumod Sundar. 2022. Analyzing the performance of sentiment analysis using BERT, DistilBERT, and RoBERTa. In *2022 IEEE international power and renewable energy conference (IPRECON)*. IEEE, 1–6.
- [20] Fei Liao. 2025. Detecting Public Perceptions of Apollo Go on Chinese Social Media Based on Topic Modeling and Sentiment analysis. In *Proceedings of the 2024 5th International Conference on Computer Science and Management Technology (ICCSMT '24)*. Association for Computing Machinery, New York, NY, USA, 61–65. doi:10.1145/3708036.3708046
- [21] Wan Liu, Zhechong Luo, and Wan Chen. 2024. Narrative Roads to Rebuild China's Global Image -Sentiment Analysis of Twitter Activities after China's Covid-19 Aid Activities in Italy. In *Proceedings of the 2023 6th International Conference on E-Business, Information Management and Computer Science* (Hong Kong, Hong Kong) (EBIMCS '23). Association for Computing Machinery, New York, NY, USA, 12–24. doi:10.1145/3644479.3644482
- [22] Nicole Lupien, Sukeshini A Grandhi, Linda Plotnick, and Star Roxanne Hiltz. 2017. Wait, Did You Say No Internet? An Exploratory Study of the Perceived Impact of Internet Outage. In *Companion of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 231–234.
- [23] Florence Madenga. 2021. From transparency to opacity: Storytelling in Zimbabwe under state surveillance and the internet shutdown. *Information, Communication & Society* 24, 3 (2021), 400–421.
- [24] M S Neethu and R Rajasree. 2013. Sentiment analysis in twitter using machine learning techniques. In *2013 Fourth International Conference on Computing, Communications and Networking Technologies (ICCCNT)*. 1–5. doi:10.1109/ICCCNT.2013.6726818
- [25] Song Yi Ng, Kian Ming Lim, Chin Poo Lee, and Jit Yan Lim. 2023. Sentiment Analysis using DistilBERT. In *2023 IEEE 11th Conference on Systems, Process & Control (ICSPC)*. 84–89. doi:10.1109/ICSPC59664.2023.10420272
- [26] Ravena Panicker. 2020. Internet Shutdown: Is It Violation of Fundamental Rights? Available at SSRN 3667818 (2020).
- [27] Rubén Pazmiño, Fernando Badillo, Miguel Conde González, and Francisco José García-Peñalvo. 2021. Ecuadorian Higher Education in COVID-19: A Sentiment Analysis. In *Eighth International Conference on Technological Ecosystems for Enhancing Multiculturality* (Salamanca, Spain) (TEEM'20). Association for Computing Machinery, New York, NY, USA, 758–764. doi:10.1145/3434780.3436679
- [28] Nusrat Jahan Prottasha, Abdullah As Sami, Md Kowsher, Saydul Akbar Murad, Anupam Kumar Bairagi, Mehedi Masud, and Mohammed Baz. 2022. Transfer learning for sentiment analysis using BERT based supervised fine-tuning. *Sensors* 22, 11 (2022), 4157.
- [29] Andrei Soldatov and Irina Borogan. 2017. *The Red Web: The Kremlin's Wars on the Internet*. Perseus Books, USA.
- [30] Ahmed Sultan, Mahmoud Salim, Amina Gaber, and Islam El Hosary. 2020. WESSA at SemEval-2020 Task 9: Code-mixed sentiment analysis using transformers. *arXiv preprint arXiv:2009.09879* (2020).
- [31] Anirudh Tagat, Amreesh Phokeer, and Hanna M. Kreitem. 2024. Net Loss: An econometric method to measure the impact of Internet shutdowns. *ACM J. Comput. Sustain. Soc.* 2, 2, Article 25 (June 2024), 22 pages. doi:10.1145/3659466
- [32] The Daily Star. 2024. *Broadband Internet Restored in Selected Areas*. <https://www.thedailystar.net/news/bangladesh/news/broadband-internet-restored-selected-areas-3660371> Accessed: 2025-01-20.
- [33] The Daily Star. 2024. *The Razakar Back-and-Forth: Who Said What?* <https://www.thedailystar.net/news/bangladesh/news/the-razakar-back-and-forth-who-said-what-3657341> Accessed: 2025-01-20.
- [34] The Daily Star. 2024. What You Need to Know About Internet Crackdown in Bangladesh. *The Daily Star* (2024). <https://www.thedailystar.net/business/news/what-you-need-know-about-internet-crackdown-bangladesh-3676346> Accessed: 2025-01-20.
- [35] Patricia Vargas-Leon. 2016. Tracking Internet shutdown practices: Democracies and hybrid regimes. In *The turn to infrastructure in Internet governance*. Springer, 167–188.
- [36] Asra Sakeen Wani. 2024. Use of ICTs during ongoing protracted socio-political disruptions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 447, 6 pages. doi:10.1145/3613905.3638194
- [37] Darrell M West. 2016. Internet shutdowns cost countries \$2.4 billion last year. *Center for Technological Innovation at Brookings, Washington, DC* (2016).
- [38] Linlin You and Bige Tunçer. 2016. Exploring public sentiments for livable places based on a crowd-calibrated sentiment analysis mechanism. In *Proceedings of the 2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining* (Davis, California) (ASONAM '16). IEEE Press, 693–700.
- [39] J. Zittrain and B. Edelman. 2003. Internet filtering in China. *IEEE Internet Computing* 7, 2 (2003), 70–77. doi:10.1109/MIC.2003.1189191
- [40] Hüseyin İlgin and Erdal Kılıç. 2021. Sentiment Analysis using Transformers and Machine Learning Models. In *2021 6th International Conference on Computer Science and Engineering (UBMK)*. 42–45. doi:10.1109/UBMK52708.2021.9558931

A APPENDIX

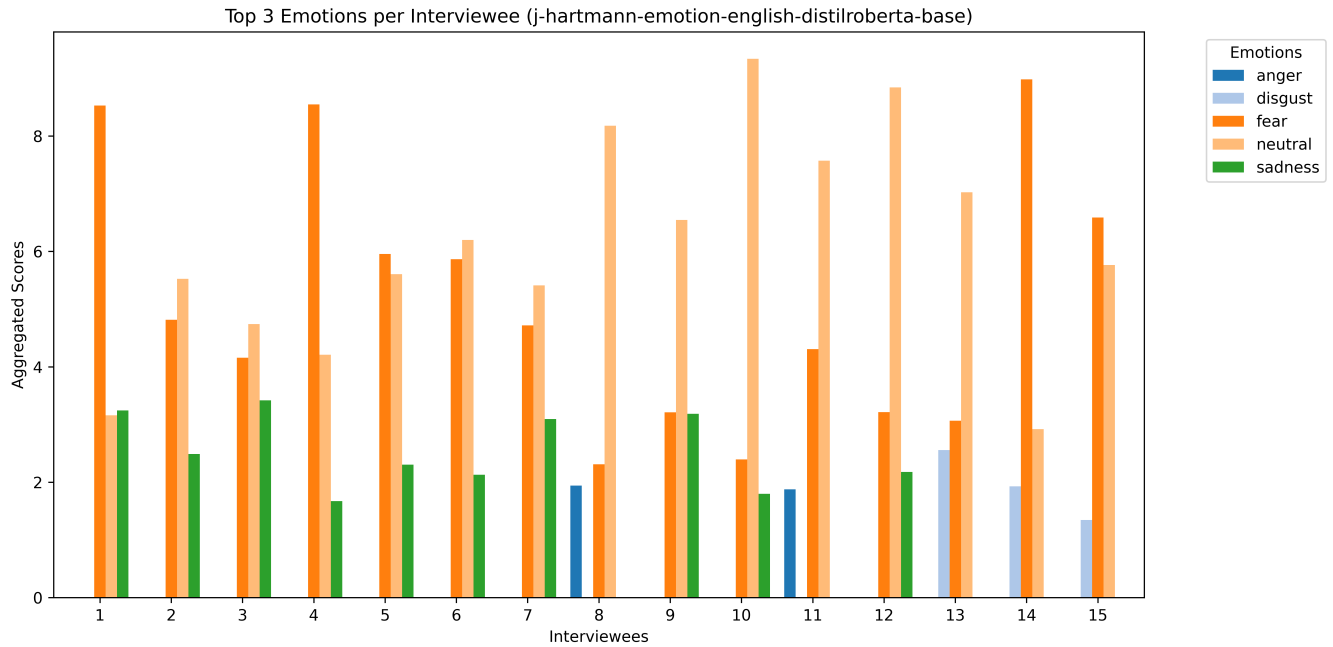


Figure 2: Sentiment Analysis of the Experiences of the Interviewees Using Model 2



Figure 3: Sentiment Analysis of the Experiences of the Interviewees Using Model 3

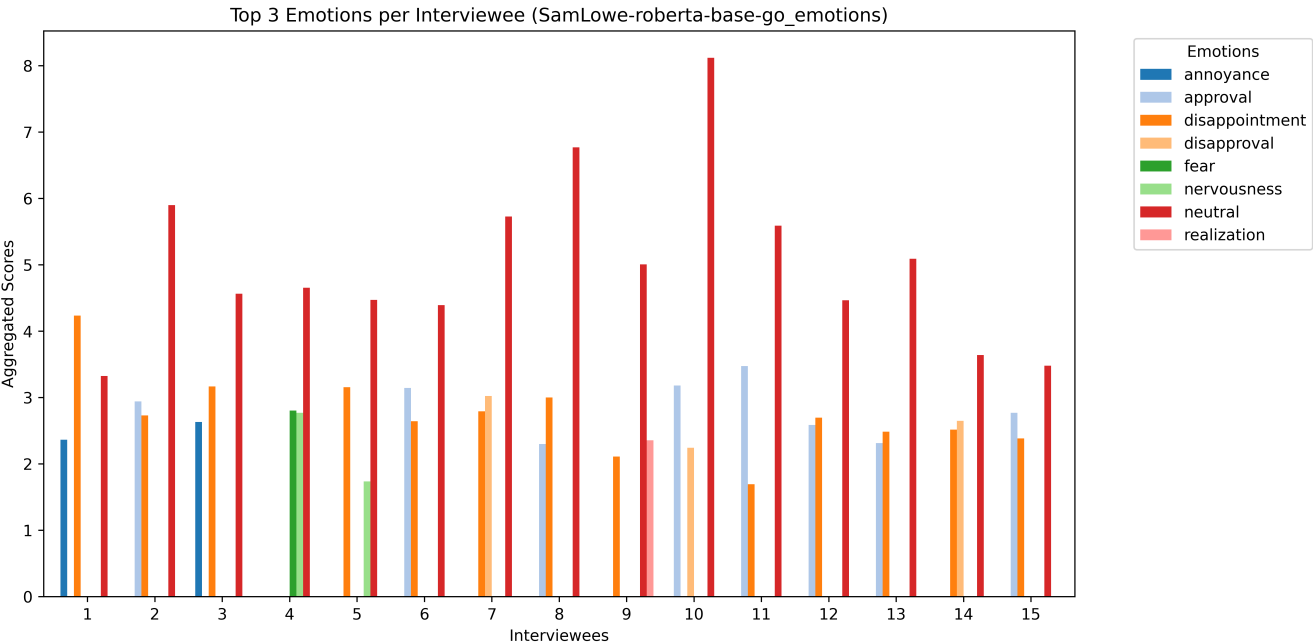


Figure 4: Sentiment Analysis of the Experiences of the Interviewees Using Model 4